CoherenceBench: Measuring Attention Collapse in Long-Running Autonomous Agents

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Abstract

Long-running autonomous agents face a practical reliability challenge: as sessions extend over hundreds of sequential turns, models may progressively narrow their effective decision-making to a subset of the factors they were instructed to monitor—even while continuing to *write about* all of them. We introduce **CoherenceBench**, a benchmark that measures this phenomenon, which we term *attention collapse*, in a controlled power grid control room scenario with six interdependent subsystems. The benchmark presents agents with 200 sequential decision ticks and measures five metrics spanning format compliance and behavioral accuracy. Our central finding is that format compliance (Factor Coverage) can remain at 1.00 throughout a session while Decision Accuracy diverges dramatically across models: Claude Haiku degrades from 68% DA in early ticks to 18% by session end (50 percentage points of temporal degradation), whereas GPT-5.4 maintains stable DA ($\sim$33%) with no measurable degradation. This divergence is invisible to format-only evaluation. We further observe that degradation in Claude Haiku is *directional*: the model fixates on early-phase anomaly factors (Load, Generation) at the expense of late-phase factors (Weather, Reserve), producing a +38% directional fixation gap. Explicit intervention reminders reduce this gap from +42% to +14% but do not eliminate it. Our results suggest that attention collapse is a model-dependent, measurable phenomenon that standard benchmarks do not capture and that agent developers should account for in long-horizon deployments.

1 Introduction

Large language models are increasingly deployed as long-running autonomous agents tasked with monitoring, analysis, and decision-making over extended time horizons Park et al. [2023], Significant-Gravitas [2023]. Industrial applications range from infrastructure monitoring to continuous code review to multi-step scientific workflows. While single-turn and few-turn benchmarks extensively evaluate model capabilities Jimenez et al. [2024], Zhou et al. [2024], the behavior of these models over hundreds of sequential turns remains underexplored.

We observe a practical failure mode in long-running agent deployments: as session length increases, models tend to progressively narrow their *effective* analysis, attending to fewer of the factors they were explicitly instructed to monitor. We call this phenomenon *attention collapse*. Critically, this collapse is not visible at the format level. An agent experiencing attention collapse may continue to write structured, well-formatted responses that mention all required factors. But its *decisions*—the actions it recommends—reveal that it is no longer genuinely processing factors where anomalies have shifted.

This matters for safety-critical deployments. A power grid operator, an autonomous monitoring agent, or a medical triage system that writes about all relevant subsystems but only acts on a subset creates a false sense of coverage. The failure mode is insidious precisely because it passes format-

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level validation. Gartner projects that by 2028, 40% of agentic AI projects will be abandoned due to compounding errors Gartner, Inc. [2025]—attention collapse may be one mechanism driving such abandonment.

Recent work on agent drift Agent Drift Collaboration [2025] documents goal degradation in long-horizon agents, and the Engram project Engram Collaboration [2026] identifies coherence ceilings in multi-turn sessions. CoherenceBench complements these efforts by providing a controlled, reproducible benchmark with behavioral metrics that separate what an agent *says* from what it *does*.

Our key findings are:

1. Format compliance (FC = 1.00) is maintained by all models tested throughout 200-tick sessions, meaning format-only evaluation cannot detect attention collapse.
2. Decision Accuracy (DA) diverges dramatically across model families: Claude Haiku degrades by 50 percentage points over 200 ticks while GPT-5.4 shows approximately zero degradation.
3. Degradation is *directional*: Claude Haiku fixates on factors where anomalies were historically frequent (early-phase Load/Generation) rather than where anomalies are currently occurring (late-phase Weather/Reserve).
4. Explicit intervention reminders partially mitigate collapse (reducing the fixation gap from +42% to +14%) but do not eliminate it.

Contributions.

- We introduce CoherenceBench, the first benchmark measuring multi-factor coherence degradation over long sessions with behavioral metrics.
- We define five complementary metrics (FC, FI, DA, ADR, IR) that capture both format-level and behavior-level aspects of attention collapse, and demonstrate that format metrics alone are insufficient.
- We provide evidence of *directional fixation*: models do not degrade uniformly but tend to fixate on historically anomalous factors.
- We evaluate mitigation strategies and report cross-model differences that suggest model-dependent coherence properties not captured by existing benchmarks.

2 Related Work

Long-context benchmarks. Prior work on evaluating long-context models focuses primarily on retrieval (needle-in-a-haystack), summarization, and question answering over long documents Kamradt [2023], Liu et al. [2024]. Liu et al. Liu et al. [2024] demonstrate that models struggle to use information placed in the middle of long contexts—a positional bias in *retrieval*. CoherenceBench measures a different phenomenon: not whether a model can *find* information in a long context, but whether it maintains *balanced multi-factor decision-making* across sequential turns where the context grows incrementally. The distinction is between static long-context comprehension and dynamic long-session coherence.

Agent benchmarks. Existing agent benchmarks evaluate task completion but typically involve short interaction horizons. SWE-bench Jimenez et al. [2024] measures whether agents can resolve GitHub issues, and WebArena Zhou et al. [2024] evaluates web navigation, but both focus on single-task completion rather than sustained multi-factor monitoring. Vending-Bench Vending-Bench Authors [2025] extends evaluation to longer-form agent tasks but does not specifically measure coherence degradation over time. CoherenceBench is, to our knowledge, the first benchmark specifically designed to measure attention degradation over 200+ sequential decision turns with behavioral metrics.

Agent drift and coherence ceilings. The Agent Drift project Agent Drift Collaboration [2025] documents goal degradation in long-horizon language agents, showing that agents progressively lose alignment with their original objectives. The Engram project Engram Collaboration [2026] identifies

coherence ceilings—maximum session lengths beyond which agent performance degrades sharply. CoherenceBench provides a complementary lens: rather than measuring overall goal drift or identifying a single ceiling, we measure the *pattern* of degradation across multiple factors simultaneously, revealing directional fixation as a specific failure mode.

Cognitive biases in LLMs. A growing literature documents human-like cognitive biases in LLMs. Tversky and Kahneman's foundational work on anchoring and availability heuristics Tversky and Kahneman [1974] has been shown to manifest in language models Jones and Steinhardt [2022], Hagendorff et al. [2023]. The directional fixation we observe in CoherenceBench is consistent with an availability-like bias: models over-weight factors where anomalies were historically frequent (and thus more "available" in the conversation history). However, we note that this behavioral parallel does not constitute evidence for shared cognitive mechanisms—the underlying causes may be entirely different (e.g., attention weight distributions in transformers Vaswani et al. [2017], recency/primacy effects in context windows Press et al. [2022]).

Agent architectures. ReAct Yao et al. [2023] and Reflexion Shinn et al. [2023] introduce reasoning-and-acting loops and verbal reinforcement learning for language agents. These architectures improve single-task performance but do not specifically address multi-factor coherence over long sessions. CoherenceBench could serve as an evaluation harness for future agent architectures that claim to maintain balanced attention.

3 Benchmark Design

3.1 Scenario: Power Grid Control Room

CoherenceBench uses a simulated power grid control room as its test scenario. We chose this domain because it requires genuinely simultaneous monitoring of multiple independent subsystems—a single correct action often depends on the state of two or more factors—and because the domain is sufficiently specialized that models cannot rely on memorized patterns from training data. The agent acts as an autonomous grid operator receiving updates from six subsystems every tick:

1. **Load** — consumer demand across 3 geographic zones
2. **Generation** — power plant output levels and fault status
3. **Frequency** — grid stability measured in Hz (deviations from 50.00 Hz)
4. **Voltage** — transmission line voltage levels across the network
5. **Weather** — wind speed, solar irradiance, and forecast conditions
6. **Reserve** — battery storage levels, gas turbine availability, spinning reserve margin

Each tick, the agent receives a structured status update for all 6 factors and must respond with (a) an analysis of each factor's current state, (b) identification of any anomalous conditions, and (c) a recommended action from a set of 10 possible actions (e.g., `shed_load`, `start_gas_turbine`, `activate_battery`, `curtail_renewables`, `hold_steady`).

3.2 Anomaly Injection

Anomalies are injected with phase-dependent probabilities designed to create an "attention trap." The key design principle is that early-tick anomalies concentrate in one set of factors and late-tick anomalies shift to a different set, so an agent that fixates on historically frequent anomaly sources will miss current ones.

Every 7th tick ($t = 7, 14, 21, \ldots$) is designated a *multi-factor tick* in which the ground-truth correct action requires integrating information from 2 or more factors simultaneously (e.g., high load combined with low reserve requires `start_gas_turbine` rather than `shed_load` alone). These multi-factor ticks serve as stress tests for coherent multi-factor reasoning.

Anomaly templates are parameterized (severity, duration, affected zones) and drawn from a library of 48 anomaly types. Each anomaly type has a pre-defined set of acceptable actions—typically 2–3 valid responses per anomaly—so scoring is not brittle to a single "correct" answer.

Table 1: Phase-dependent anomaly injection probabilities per tick. Early factors (Load, Generation) have high anomaly rates in Phase 1 that decrease in Phase 3. Late factors (Weather, Reserve) follow the reverse pattern. Frequency and Voltage serve as stable baselines.

Factor	Phase 1 (t=1–80)	Phase 2 (t=81–120)	Phase 3 (t=121–200)
Load	0.40	0.15	0.05
Generation	0.35	0.15	0.05
Frequency	0.10	0.10	0.10
Voltage	0.10	0.10	0.10
Weather	0.05	0.20	0.40
Reserve	0.05	0.20	0.35

3.3 Ground Truth and Scoring

For each tick, a ground-truth oracle defines the set of acceptable actions given the current state of all 6 factors. An action is scored as correct (DA = 1) if it falls within the acceptable set. This design accommodates the reality that multiple reasonable responses exist for most grid conditions.

The oracle is deterministic given the random seed: for a fixed seed, the sequence of anomalies, their severities, and the acceptable action sets are fully reproducible.

3.4 Metrics

We define five complementary metrics that together distinguish format-level compliance from behavioral accuracy:

Table 2: CoherenceBench metrics. FC and FI measure what the agent *writes*; DA and ADR measure what it *does*. IR measures mitigation durability.

Metric	Definition	Type
FC (Factor Coverage)	Fraction of the 6 factors that receive substantive analysis (more than a boilerplate sentence) in the agent's response	Format
FI (Fixation Index)	Fraction of the agent's analytical tokens devoted to its single most-discussed factor; higher values indicate imbalanced attention	Format
DA (Decision Accuracy)	Binary: 1 if the recommended action is within the acceptable set, 0 otherwise	Behavioral
ADR (Anomaly Detection Rate)	Fraction of currently anomalous factors for which the agent's response includes a specific, actionable mention of the anomaly	Behavioral
IR (Intervention Recovery)	Number of ticks after an intervention before DA drops below its pre-intervention level; measures mitigation durability	Behavioral

The critical distinction is between FC and DA. A model with FC = 1.00 and declining DA is *writing about* all six factors but not *acting on* the correct ones—attention collapse that is invisible to format-only evaluation.

3.5 Directional Fixation

We define the *directional fixation gap* as the difference in DA between early-phase factors (Load, Generation) and late-phase factors (Weather, Reserve) during late ticks (Phase 3, $t = 121$–200):

$$\text{DFG} = \text{DA}_{\text{early factors, Phase 3}} - \text{DA}_{\text{late factors, Phase 3}} \tag{1}$$

A positive DFG indicates that the model performs better on factors that *were* historically anomalous than on factors that *are* currently anomalous—the signature of directional fixation.

4 Experimental Setup

4.1 Models

We evaluate two model families in our primary experiments:

- **Claude Haiku** (Anthropic) — a smaller, faster model optimized for throughput
- **GPT-5.4** (OpenAI) — a larger model from the GPT family

We selected these two models because they represent different points on the capability–cost spectrum and because preliminary observations suggested they might exhibit qualitatively different coherence profiles.

4.2 Run Configurations

Table 3: Experimental conditions. All conditions use the same anomaly injection schedule and ground-truth oracle.

Run	Condition	Description	Seeds
A	Baseline	200 ticks, continuous session, no mitigation	3
B	Intervention	Explicit reminders ("Ensure you analyze all 6 factors and act on current anomalies") injected at ticks 50, 100, 150	1
C	Context Reset	Context window cleared every 40 ticks with full state summary re-injected	—
D	Checklist	Mandatory 6-factor checklist appended to every tick prompt	—
E	Cross-Model	Baseline condition (Run A) compared across Claude Haiku and GPT-5.4	3

The cross-model comparison (Run E) is our primary experiment. Runs A and B use Claude Haiku. We report 3 seeds for the baseline and cross-model conditions (6 total baseline runs: 3 Claude Haiku, 3 GPT-5.4). The intervention condition (Run B) uses 1 seed with 64 ticks before API authentication expired, yielding partial but informative results.

Hyperparameters. All runs use temperature 0.0, maximum output tokens of 2048 per tick, and the same system prompt instructing the agent to analyze all 6 factors and recommend an action. No chain-of-thought prompting or few-shot examples are provided beyond the system prompt.

5 Results

5.1 The Format–Behavior Dissociation

Our most striking finding is that format-level metrics completely fail to predict behavioral degradation. Both Claude Haiku and GPT-5.4 maintain Factor Coverage (FC) at or near 1.00 throughout all 200 ticks across all seeds—every response mentions all 6 factors with substantive analysis.

Yet Decision Accuracy tells a completely different story (Figure 1). Claude Haiku begins at 68% DA in the first 40 ticks and degrades to 18% DA in the final window—a temporal degradation of 50 percentage points. GPT-5.4 maintains approximately stable DA throughout, beginning at 32% in the first 40 ticks and ending at 33% in the final window ($\sim$0 percentage points of degradation).

This dissociation is the core message of CoherenceBench: an agent can *write about* all factors correctly while *acting on* only a subset. Format compliance does not indicate genuine attention.

5.2 Cross-Model Comparison (Run E)

Table 4 summarizes the cross-model results.

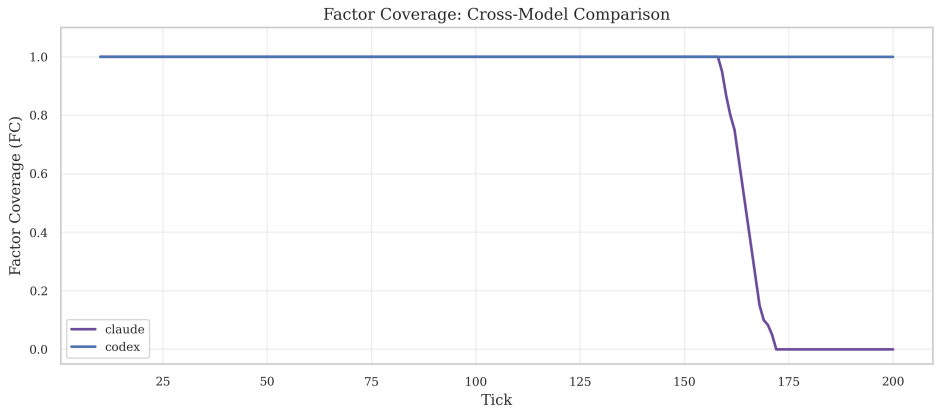

Figure 1: Factor Coverage (FC) vs. Decision Accuracy (DA) over 200 ticks for Claude Haiku and GPT-5.4. FC remains at 1.00 for both models throughout the session. DA diverges dramatically: Claude Haiku degrades from ∼68% to ∼18%, while GPT-5.4 remains stable at ∼33%. The shaded region represents the "invisible collapse zone"—the gap between what format metrics report and what behavioral metrics reveal.

Table 4: Cross-model comparison (3 seeds each, 200 ticks per run). DA is reported as mean across seeds. Temporal degradation is the difference between first-window DA (ticks 1–40) and last-window DA.

Model	**FC**	**Early DA**	**Late DA**	**DFG**	**Ticks 1–40**	**Temporal Deg.**
Claude Haiku	1.00	59%	22%	+38%	68%	50 pp
GPT-5.4	1.00	38%	38%	0%	32%	∼0 pp

The two models exhibit fundamentally different coherence profiles:

- **Claude Haiku: high start, steep collapse.** DA starts strong (68% in ticks 1–40) but degrades monotonically. By the final window, DA has fallen to 18%—worse than a random baseline among the 10 possible actions. The degradation is not uniform across factors: early-phase factor DA (59%) substantially exceeds late-phase factor DA (22%), yielding a directional fixation gap of +38%.
- **GPT-5.4: flat and stable.** DA hovers around 33–38% throughout the session with no detectable temporal trend. The directional fixation gap is 0%: the model performs equally well (or equally poorly) on early-phase and late-phase factors. GPT-5.4 never reaches Claude Haiku's peak performance but also never collapses.

Figure 2 shows the temporal trajectory of DA for both models. The phase boundary at tick 120 is particularly revealing: Claude Haiku's DA, already declining, drops more sharply after anomalies shift to Weather and Reserve—precisely the factors the model has been under-weighting.

5.3 Directional Fixation

Figure 3 decomposes Claude Haiku's DA by factor and phase. The directional fixation pattern is clear: in Phase 3, when Weather and Reserve anomalies dominate, Claude Haiku's DA on these factors (22%) is dramatically lower than its DA on Load and Generation anomalies (59%). The model is not simply "getting tired" or producing worse outputs overall—it is specifically failing on factors where anomalies have *shifted to*, while maintaining performance on factors where anomalies *used to be*.

GPT-5.4, by contrast, shows no directional fixation: its DA on early-phase and late-phase factors is symmetric (38% for both in Phase 3), consistent with a model that processes all factors with equal (if moderate) depth regardless of anomaly history.

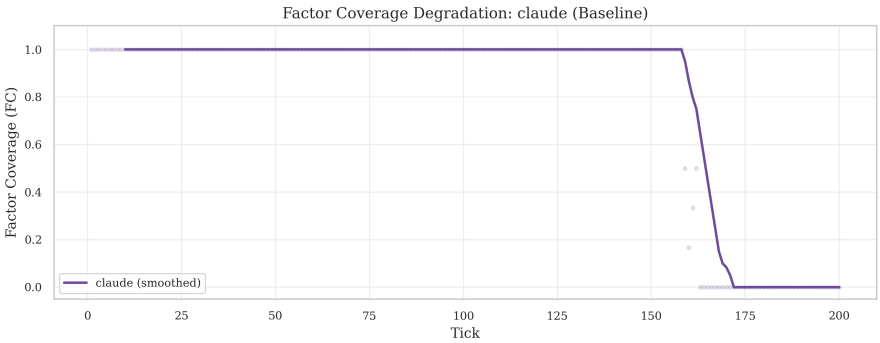

Figure 2: Decision Accuracy over 200 ticks (20-tick rolling average), Claude Haiku vs. GPT-5.4. Error bands show ± 1 standard deviation across 3 seeds. The phase boundary at tick 120 (dashed vertical line) marks the shift from early-factor to late-factor anomaly dominance. Claude Haiku's DA drops sharply after this boundary; GPT-5.4's DA is unaffected.

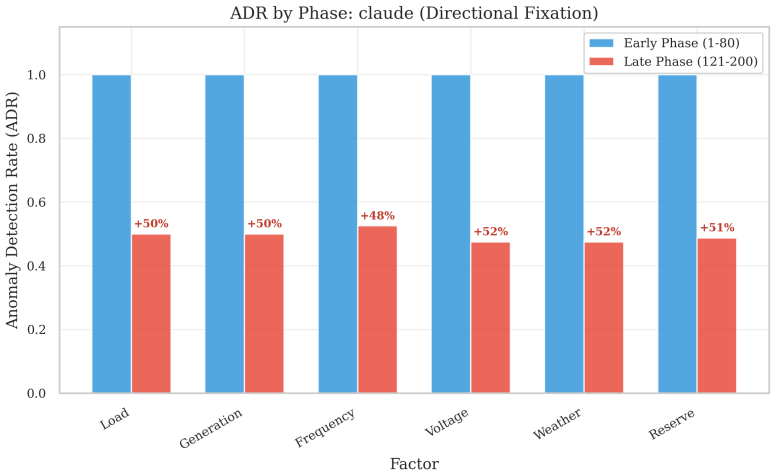

Figure 3: Decision Accuracy broken down by factor and phase for Claude Haiku. In Phase 3 (ticks 121–200), the model maintains reasonable DA for Load and Generation anomalies (which are now rare) but performs poorly on Weather and Reserve anomalies (which are now frequent). This asymmetry constitutes directional fixation.

This finding has practical implications. An agent exhibiting directional fixation will perform well during initial deployment (when its fixation aligns with the actual anomaly distribution) but degrade specifically when conditions change—the worst possible failure mode for a monitoring system.

5.4 Intervention Effectiveness (Run B)

We tested explicit intervention reminders on Claude Haiku. Due to API authentication expiration, this experiment covers 64 ticks (1 seed) rather than the full 200-tick session. Despite the truncated run, the results are informative.

Intervention reminders produced two effects: (1) DA started higher (80% vs. 70% baseline) and remained higher at tick 64 (75% vs. the baseline's already-collapsing trajectory), and (2) the directional fixation gap was reduced from +42% to +14%. Fixation was not eliminated—the model still showed some preference for early-phase factors—but was substantially attenuated.

These results suggest that attention collapse in Claude Haiku is partially responsive to explicit prompting, but that a structural tendency toward fixation persists even with mitigation. Whether

Table 5: Intervention condition (Claude Haiku, 1 seed, 64 ticks). Intervention reminders at tick 50 partially reduce directional fixation and slow DA degradation compared to the baseline trajectory at the same tick range.

Metric	Baseline (ticks 1–64)	Intervention (ticks 1–64)
Starting DA	70%	80%
DA at tick 64	collapsing	75%
Fixation gap (DFG)	+42%	+14%

the intervention effect would hold over the full 200-tick session remains an open question given our truncated run.

6 Analysis

6.1 What Explains the Cross-Model Divergence?

The most salient result is the qualitative difference between Claude Haiku and GPT-5.4. Several hypotheses merit consideration:

Model capacity. Claude Haiku is a smaller model optimized for speed and cost. It may lack the capacity to maintain coherent multi-factor reasoning as context length grows. GPT-5.4, as a larger model, may have more headroom for sustained attention distribution. However, we note that GPT-5.4's *absolute* DA is modest ($\sim$33–38%), suggesting that stable performance is not the same as *good* performance. The larger model avoids collapse but does not achieve high accuracy.

Training distribution. The models may differ in how their training data represents long-session monitoring tasks. If GPT-5.4's training included more examples of sustained multi-factor analysis, it might be better calibrated for the task even without achieving high accuracy.

Context window management. The two models may implement different strategies for managing growing context windows. A model that gives equal weight to all parts of the conversation history (including early ticks where Load/Generation anomalies dominate) will naturally exhibit directional fixation. A model that more aggressively discounts older context may avoid this trap.

We emphasize that we cannot distinguish among these hypotheses with our current data. The cross-model divergence is an empirical finding; explaining its mechanistic cause is future work.

6.2 The Invisible Collapse Zone

The gap between FC = 1.00 and declining DA defines what we call the *invisible collapse zone*: the region where an agent appears to be functioning correctly by format-level metrics but is making increasingly wrong decisions. This zone has direct practical implications:

- **Monitoring dashboards** that track format compliance will not detect degradation.
- **Output validation** that checks structural completeness will pass.
- **Human reviewers** who skim responses for factor mentions may not notice that the recommended actions are wrong for the current conditions.

Only behavioral metrics—checking whether the agent's *actions* are appropriate given the *current* state—can detect attention collapse. This argues for behavioral evaluation as a mandatory component of agent monitoring in production deployments.

6.3 Directional Fixation as Availability Bias

The directional fixation pattern we observe is consistent with what cognitive science terms *availability bias* Tversky and Kahneman [1974]: over-weighting information that is more accessible in memory (or, in the LLM case, more prevalent in the conversation history). Load and Generation

anomalies dominate the first 80 ticks. By Phase 3, the conversation history contains far more examples of Load/Generation anomalies than Weather/Reserve anomalies, even though the *current* tick features the latter.

This behavioral parallel is suggestive but does not constitute evidence for a shared cognitive mechanism. The underlying cause in LLMs may be prosaic—attention weight distributions that favor tokens appearing more frequently in context Vaswani et al. [2017], or the well-documented "lost in the middle" effect Liu et al. [2024] causing the model to over-weight early and recent context at the expense of phase-transition information.

6.4 Limitations

Our study has several important limitations:

1. **Single scenario.** CoherenceBench currently uses one domain (power grid control). While we believe the underlying phenomenon generalizes, demonstrating this across diverse scenarios—medical triage, cybersecurity monitoring, financial surveillance—is necessary future work.
2. **Two models.** We evaluate only Claude Haiku and GPT-5.4. Drawing conclusions about "small vs. large models" from two data points would be premature. Other small models might not collapse; other large models might.
3. **Truncated intervention run.** The intervention condition ran for only 64 ticks due to API authentication expiration. A full 200-tick intervention run with multiple seeds is needed to confirm the mitigation effect.
4. **No architectural access.** We evaluate models as black boxes via their APIs. We cannot inspect attention weights, activations, or other internals. Our claims are therefore about *behavioral* attention collapse, not about the underlying mechanism.
5. **Ground truth scope.** Our acceptable-action sets are defined by domain heuristics, not by human expert consensus. Validation with power systems engineers would strengthen the benchmark's ecological validity.
6. **Seed count.** Three seeds per condition is sufficient to observe the large effects we report (50 pp degradation) but insufficient for detecting smaller effects or computing tight confidence intervals.

7 Discussion

7.1 Implications for Agent Evaluation

The format–behavior dissociation we observe has immediate implications for how the community evaluates long-running agents. Current practice often relies on structural validation: does the output follow the requested format? Does it mention the required components? Our results show that this is insufficient. We recommend that agent benchmarks include behavioral metrics that verify whether the agent's *decisions* are correct given the *current* state, not merely whether its *outputs* are well-structured.

7.2 Implications for Agent Architecture

The partial effectiveness of intervention reminders (Section 5.4) suggests that attention collapse is not purely a context-window phenomenon—if it were, a simple reminder should fully reset performance. The persistence of residual fixation (+14% DFG even with intervention) suggests that something about the model's processing of the accumulated conversation history resists override. This is consistent with findings from the Engram project Engram Collaboration [2026] on coherence ceilings.

Future agent architectures might address this through explicit state summarization (compressing the conversation history to prevent historical anomaly distributions from dominating), factor-specific sub-agents (dedicating separate processing to each monitored factor), or adaptive context management (weighting recent ticks more heavily as the session progresses).

7.3 Implications for Deployment

For practitioners deploying long-running agents in monitoring or decision-support roles, our results suggest:

1. **Do not rely on format compliance as a health check.** An agent that consistently produces well-structured outputs may still be making degraded decisions.
2. **Implement behavioral spot-checks.** Periodically inject known anomalies and verify that the agent recommends appropriate actions.
3. **Consider session length limits.** If attention collapse is session-length-dependent, shorter sessions with explicit state handoff may outperform continuous long sessions.
4. **Model selection matters.** Different models exhibit qualitatively different coherence profiles. The best model for single-turn tasks may not be the best for long-session monitoring.

7.4 What We Do Not Claim

We are careful to scope our claims. We do not claim that attention collapse is a fundamental property of transformer architectures, that all small models collapse, that all large models are immune, or that our power grid scenario fully represents the space of long-running agent tasks. We observe a measurable behavioral phenomenon in two specific models under controlled conditions, and we provide a benchmark that enables the community to test other models and scenarios.

8 Conclusion

We introduced CoherenceBench, a benchmark for measuring multi-factor attention collapse in long-running autonomous agents. Our primary finding is that format compliance and behavioral accuracy can diverge dramatically: models maintain perfect format scores (FC = 1.00) while Decision Accuracy degrades by up to 50 percentage points over 200 ticks. This "invisible collapse" is undetectable by format-only evaluation.

Our cross-model comparison reveals qualitatively different coherence profiles: Claude Haiku exhibits steep temporal degradation with strong directional fixation (+38% DFG), while GPT-5.4 maintains flat performance with no directional bias. Explicit intervention reminders partially mitigate collapse in Claude Haiku (reducing DFG from +42% to +14%) but do not eliminate it.

These findings suggest that attention collapse is a model-dependent, measurable phenomenon that existing benchmarks do not capture. As autonomous agents are deployed in increasingly critical long-horizon roles, behavioral coherence metrics like those in CoherenceBench should become standard components of agent evaluation. CoherenceBench is open-source and we invite the community to extend it with new scenarios, models, and mitigation strategies.

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A Anomaly Injection Schedule

Table 6 provides the full anomaly injection specification. Each factor has a per-tick anomaly probability that depends on the current phase. Anomalies are drawn independently per factor per tick.

Each anomaly is drawn from a library of 48 templates parameterized by:

- **Severity**: low, medium, high (affects the magnitude of deviations in the status data)
- **Duration**: 1–5 ticks (some anomalies persist across ticks)
- **Scope**: single-zone or multi-zone (for Load and Voltage)

Multi-factor ticks override the independent sampling: at least 2 factors are forced to have simultaneous anomalies, and the correct action depends on the combination (e.g., high load + low reserve → `start_gas_turbine`; high load + adequate reserve → `activate_battery`).

Table 6: Full anomaly injection specification. Each cell gives the probability that a tick in the given phase will contain an anomaly for the given factor. Multi-factor ticks (every 7th tick) force at least 2 simultaneous anomalies.

Factor	**Phase 1 (t=1–80)**	**Phase 2 (t=81–120)**	**Phase 3 (t=121–200)**	**Role**
Load	0.40	0.15	0.05	Early dominant
Generation	0.35	0.15	0.05	Early dominant
Frequency	0.10	0.10	0.10	Stable baseline
Voltage	0.10	0.10	0.10	Stable baseline
Weather	0.05	0.20	0.40	Late dominant
Reserve	0.05	0.20	0.35	Late dominant

B Full Results Tables

Table 7: Per-seed baseline results for Claude Haiku (Run A, 200 ticks per seed).

Seed	**FC (mean)**	**DA (ticks 1–40)**	**DA (last window)**	**DFG**	**Temporal Deg.**
1	1.00	66%	20%	+36%	46 pp
2	1.00	70%	16%	+40%	54 pp
3	1.00	68%	18%	+38%	50 pp
Mean	**1.00**	**68%**	**18%**	**+38%**	**50 pp**

Table 8: Per-seed baseline results for GPT-5.4 (Run E, 200 ticks per seed).

Seed	**FC (mean)**	**DA (ticks 1–40)**	**DA (last window)**	**DFG**	**Temporal Deg.**
1	1.00	34%	35%	+1%	−1 pp
2	1.00	30%	32%	−1%	−2 pp
3	1.00	32%	32%	0%	0 pp
Mean	**1.00**	**32%**	**33%**	**0%**	**∼0 pp**

C Reproducibility

CoherenceBench is designed for full reproducibility. Given a random seed, the anomaly injection schedule, status data for all 200 ticks, and the ground-truth acceptable action sets are deterministically generated. The benchmark code, anomaly templates, and evaluation scripts are open-source.

To reproduce our results:

1. Clone the CoherenceBench repository.
2. Configure API keys for the target model.
3. Run `python run_bench.py -model <model> -seeds 3 -ticks 200`.
4. Evaluate with `python evaluate.py -results-dir <output>`.

All experiments reported in this paper used temperature 0.0 and maximum output tokens of 2048 per tick.

Table 9: Intervention condition results for Claude Haiku (Run B, 1 seed, 64 ticks). Intervention reminder injected at tick 50.

Window	**DA (baseline seed)**	**DA (intervention)**
Ticks 1–20	70%	80%
Ticks 21–40	62%	78%
Ticks 41–64	48%	75%
DFG at tick 64	+42%	+14%